MATERIAL TESTING LAB PROJECT

COURSE CODE: NCEC102

INDIAN INSTITUTE OF TECHNOLOGY
(INDIAN SCHOOL OF MINES), DHANBAD

DEPARTMENT OF CIVIL ENGINEERING

SESSION: 2024-25

TOPIC: Dry Sieve Analysis for Iron Ore

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Date of Submission:
6th November, 2024

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Introduction & objective

Sieve analysis is crucial in civil engineering to determine the particle size distribution of aggregates. This method ensures the quality and durability of materials used in construction. It provides reliable and cost-effective results. By understanding particle size distribution, engineers can optimize mix designs for concrete and asphalt, ensuring structures meet required specifications. The primary objective of this report is to perform a detailed dry sieve analysis on a sample of aggregates, following the procedures outlined in IS 2720 (Part-4): 1985. This includes collecting and preparing the sample, conducting the sieve analysis, recording observations, and interpreting the results. The aim is to determine the particle size distribution, assess the quality of the aggregates, and ensure compliance with relevant standards.

Indian standards

IS 2720 (Part-4): 1985 (Reaffirmed 2020) provides the methods for grain size analysis of soils. This standard is crucial in ensuring the quality and consistency of soil used in construction projects. It outlines two primary methods for determining the grain size distribution of soils larger than 75 microns: wet sieving and dry sieving. Wet sieving is applicable to all soil types, while dry sieving is recommended for soils with low clay content. For particles smaller than 75 microns, the standard specifies the use of the pipette method as the primary technique, with the hydrometer method as an alternative. Adherence to these methods ensures accurate and reliable results, contributing to the structural integrity and durability of construction materials. This project report follows the procedures outlined in IS 2720 (Part-4) to perform a comprehensive dry sieve analysis, ensuring compliance with industry standards.

Apparatus

To conduct a comprehensive dry sieve analysis, the following apparatus is essential:

1. **Set of Sieves**: Two sets, including sieves with mesh sizes of 100 mm, 63 mm, 20 mm, 10 mm, 4.75 mm for larger aggregates, and 2 mm, 1 mm, 600 µm, 425 µm, 300 µm, 150 µm, and 75 µm for finer particles, complete with a lid and pan.
2. **Weighing Balance**: Precision balance with an accuracy of 0.1 g for accurate measurement of aggregate weights.
3. **Mechanical Sieve Shaker**: An apparatus to facilitate the efficient and consistent shaking of sieves.
4. **Trays and Brushes**: Used for handling and cleaning the samples to ensure no particles are lost during the process.

These tools are crucial for ensuring accurate and reliable results in the particle size distribution analysis, adhering to the standards set by IS 2720 (Part-4): 1985 (Reaffirmed 2020).

procedure

- **Sample Preparation**:
  - Collect a representative soil sample from the field.
  - Dry the sample in an oven to remove moisture.
- **Sieving Process**:
  - Weigh 1000g of the dried sample.
  - Sieve the material through a 4.75mm sieve to separate gravel and sand fractions.
  - Further sieve the retained material using a set of sieves (100mm, 63mm, 20mm, 10mm, 4.75mm) and record the mass retained on each sieve.
  - Sieve the material passing through the 4.75mm sieve through finer sieves (2mm, 1mm, 600µm, 425µm, 300µm, 150µm, 75µm) and record the mass retained on each sieve.

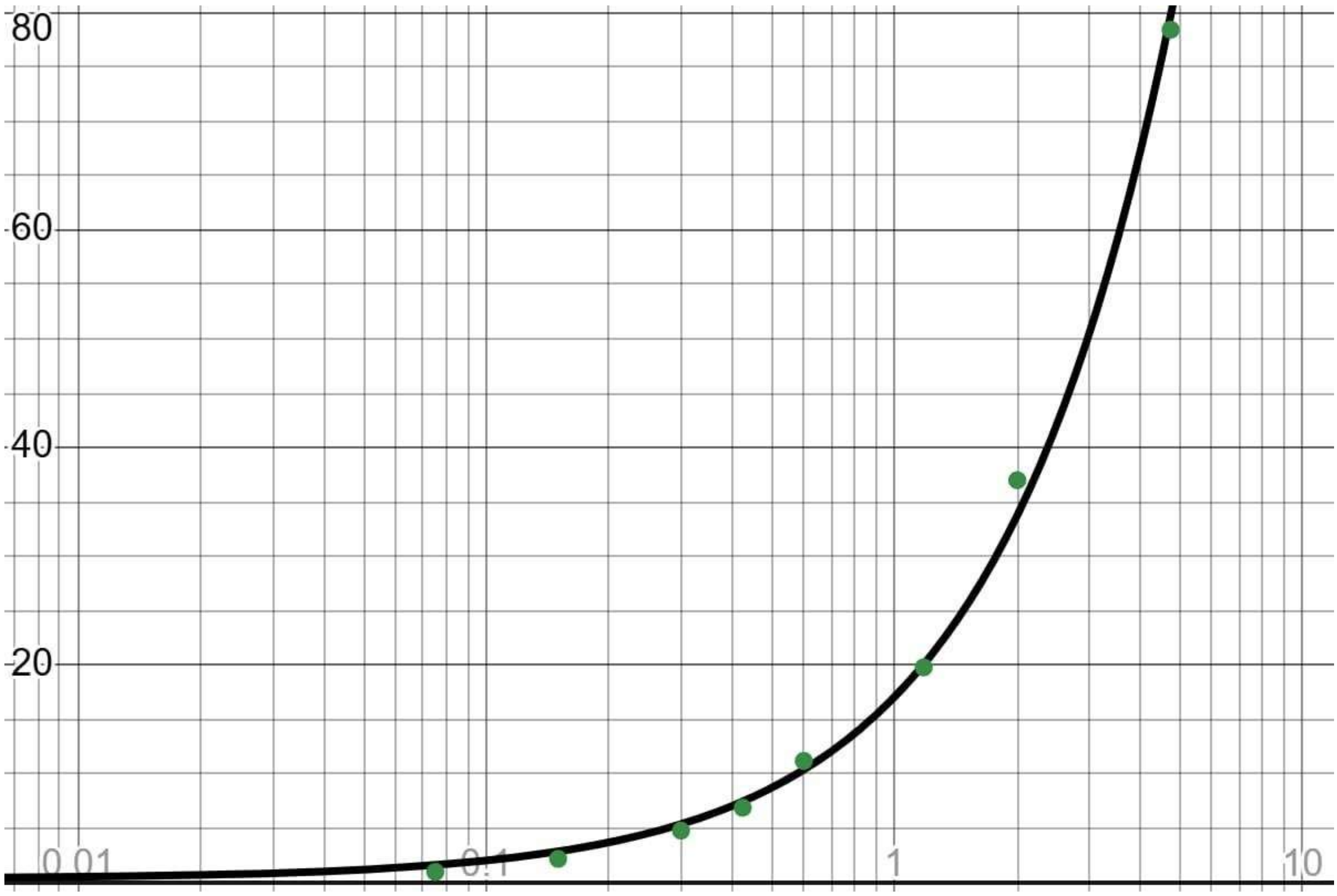
observations

Type of Material	Iron ore	Date of Testing	28.10.24
Sieving Duration	5 minutes	Tested By	Group 2
Weight of Material	1000gm	Sample No.	

S.No.	IS Sieves	Particle size D (mm)	Mass Retained (g)	Percentage Retained (%)	Cumulative %age Retained	%Finer
1	4.75mm	4.75	108	21.6	21.6	78.4
2	2mm	2	186	37.2	58.8	41.2
3	1.18mm	1.18	107	21.4	80.2	19.8
4	600 µ	0.6	43	8.6	88.8	11.2
5	425 µ	0.425	29	5.8	94.6	5.4
6	300 µ	0.3	3	0.6	95.2	4.8
7	150 µ	0.15	13	2.6	97.8	2.2
8	75 µ	0.075	6	1.2	99	1
9	Pan		5	1	100	0

Results & graph

Particle Size at 10% finer,	**D_{10}**	0.5
Particle Size at 30% finer,	**D_{30}**	1.8
Particle Size at 60% finer,	**D_{60}**	3.6
Uniformity Coefficient (Cu) =	**D60/D10**	7.2
Uniformity Curvature (Cc) =	**(D30)^2/D10 *X D60***	1.8



interpretation

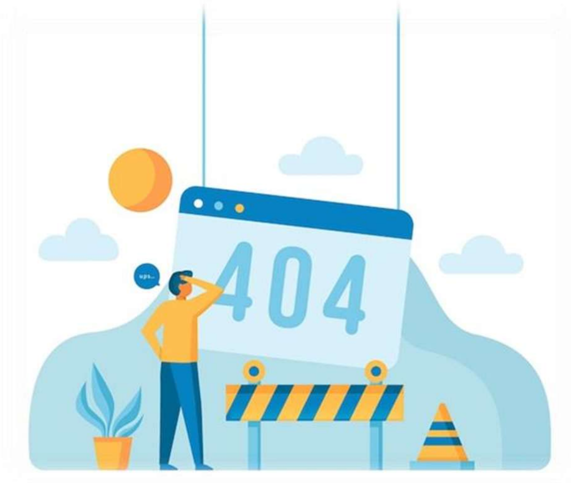
The results of the sieve analysis provide valuable insights into the particle size distribution of the tested soil sample. The percentage retained on each sieve indicates the proportion of material within specific size ranges. By analyzing the cumulative percentage retained and the percentage passing, we can determine the gradation of the aggregate. A wellgraded aggregate, with a smooth distribution curve, ensures better compaction and stability in construction applications. Conversely, poorly graded aggregates, indicated by abrupt changes in the curve, can lead to issues like segregation and reduced load-bearing capacity. Understanding these results helps in selecting appropriate materials for various construction projects, ensuring optimal performance and durability. The sieve analysis thus plays a critical role in quality control and material selection in the construction industry.

applications

Sieve analysis is widely used across various industries to ensure the quality of materials used in construction and manufacturing. In civil engineering, it helps in selecting aggregates that provide optimal strength and durability for concrete, asphalt, and road bases. The construction industry relies on sieve analysis to maintain consistency in material grading, ensuring that structures are safe and durable. Additionally, the mining industry uses this method to separate and classify mineral particles, enhancing the efficiency of extraction processes. Environmental engineering benefits from sieve analysis in soil studies, assessing soil texture, and permeability, crucial for designing foundations and drainage systems. By providing detailed insights into particle size distribution, sieve analysis supports quality control, regulatory compliance, and material optimization across multiple sectors.

limitations

While sieve analysis is a widely used method for particle size distribution, it has several limitations. Firstly, it is less effective for materials with a significant number of fines (particles smaller than 75 microns) because these particles can pass through the sieves and may not be accurately measured. Additionally, the method relies on mechanical agitation, which may not perfectly replicate the natural settling of particles. Moisture content in the sample can also affect the accuracy of the results, as clumping can alter particle distribution. Moreover, sieve analysis provides only a snapshot of particle size distribution and does not account for particle shape, which can influence material properties. Despite these limitations, sieve analysis remains a valuable tool in construction and engineering, providing essential data for material quality assessment and control.



Improvements & recommendations

To enhance the accuracy and reliability of sieve analysis, several improvements can be implemented. Firstly, ensuring the sample is thoroughly dried and free from moisture will prevent clumping and provide more accurate results. Using automated sieve shakers with consistent vibration patterns can help in achieving uniform sieving, reducing human error. Additionally, incorporating image analysis techniques alongside traditional sieve methods can give a more comprehensive understanding of particle shape and size distribution. Regular calibration of sieves and weighing

balances is essential to maintain precision. Training personnel on the correct procedures and handling of equipment can further improve the reliability of the results. By adopting these practices, the overall quality and efficiency of sieve analysis can be significantly enhanced, leading to better material selection and quality control in construction projects.

precautions

To ensure the accuracy and reliability of sieve analysis, follow these precautions:

1. **Dry Sample Thoroughly**: Ensure the sample is fully dried to prevent clumping.
2. **Clean and Calibrated Sieves**: Use sieves that are clean and properly calibrated for consistent results.
3. **Careful Handling**: Handle sieves and samples carefully to avoid material loss.
4. **Use Mechanical Sieve Shaker**: Employ a mechanical sieve shaker for uniform vibration.
5. **Wear Safety Gear**: Protect yourself with gloves and safety glasses to guard against dust and debris.
6. **Accurate Documentation**: Document all observations and measurements precisely.
7. **Regular Equipment Inspection**: Regularly inspect and maintain equipment to keep it in good working condition.
8. **Standardized Procedure**: Follow a standardized procedure to minimize human error.
9. **Check for Residual Particles**: Ensure no particles are left on the sieve by brushing them off gently.
10. **Work in a Controlled Environment**: Conduct the analysis in a dust-free, controlled environment.

conclusions

The sieve analysis provided key insights into the particle size distribution of the soil sample. The results showed a well-graded aggregate, ideal for construction applications, ensuring optimal compaction and stability. Adhering to standardized procedures is essential for quality control in material selection. Despite some limitations, sieve analysis remains crucial in civil engineering, contributing to the safety and durability of construction projects. The study emphasizes the need for continuous improvements and adherence to best practices to enhance accuracy and reliability.